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A Black Belt Consults

This section consists of a scenario and a number of exercises which will count towards your assessment.

In total the tasks in this document and the statistical exercises are worth 50% of the total marks available.

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Introduction and Case Study

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Introduction

Following a satisfactory presentation to Alan Forden the head of CS&F you have been asked to take up a Black Belt consultancy role with the company.

CS&F are at the start of their Lean 6 Sigma journey and desperately need to get started.

As you travel down for your first day you review some information that Alan's assistant has thoughtfully sent you ….

Background

Consolidated Sugar & Fats (CS&F) is a sweet manufacturer based in the small market town of Minton Blizzard. Although the company was originally privately owned it has been a limited company since the mid-1990s. The company's main brand is "Uncle Jim's" which has a worldwide appeal partly due to its nostalgic associations but also because of the excellent quality of the sweets.

In this case study we are going to limit ourselves to a particular type of confectionary known as "boiled sweets".

CS&F have 3 brands of boiled sweet which you need to know about:

"Uncle Jims" Brand

The original product upon which the company was founded. First made by hand in the 1920s the brand has the biggest profit margin, is unapologetically nostalgic, and sells world-wide.

"Dibbers Supermarket"

CS&F have a contractual obligation to provide Dibbers with their "Own Brand" boiled sweets.

"Target" Brand

Devised by CS&F as a way to help mitigate their fixed costs. When there is no need to produce the other brands then the production line makes Target. Value sweets, priced to sell, CS&F do not guarantee supply of these sweets.

A short overview of the boiled sweet manufacturing process

CS&F have two production lines that manufacture boiled sweets. The original process was semi-automated in the late 1970s and has been maintained ever since. This process is crude by modern standards but does give an “old fashioned” feel to the sweets made due to the variance of the sweet forming process. The line consists of a number of independent units. Some have been replaced with more modern models over the years but the sweet formation, which gives the sweets their old-fashioned look and feel, remains unchanged.

In your talks with the company you have discovered that they appear to be concerned especially about the quality of the “wrapping station” (the unit that wraps each sweet in a plastic film). Uncle Jim’s uses only natural ingredients their confectionery, each sweet consists mainly of sugar with a small percentage of flavourings added.

The sugar and other ingredients are melted in a boiler unit (boiler) which is positioned over a moving belt. The boilers are sealed units similar to pressure cookers. As the mixture heats up it melts and the internal pressure builds up. The pressure forces the viscous mixture through a nozzle which has a number of holes. The pressure forces the molten mixture through the holes in discrete drips. The falling drips impact the belt which results in the characteristic “boiled sweet” shape.

The sweets then flow under a cooling fan and then pass into an automatic wrapping device which wraps them in a plastic film.

The wrapped sweets are collected into a hopper which has a funnel at the bottom with an automatic dispensing valve.

Opened bags travel under the valve which feeds a pre-set weight (nominally 100) of sweets into each bag. The bags then travel to a sealing station which closes the bags. The bags are then auto packed into cartons of 60 bags and the cartons are sealed ready for shipping.

The factory has two production lines; this means that the company can produce two types of sweet at a time. To manufacture another type of sweet the factory needs to stop a production line and thoroughly clean it to remove any trace of the previous ingredients - this takes about 2 hours so company policy is to manufacture in large batches (typically they produce enough of one sweet to satisfy demand for 3 months).

Major Business Units

CS&F has the following major units in the business:

Unit	Value add
Purchasing	Negotiates and orders ingredients for sweet creation (sugar and flavourings)
Goods inward	Stores supplies when they arrive transfers them to Manufacturing on a weekly basis
Manufacturing	Produces the actual sweets. Bags and boxes them
Goods outbound	Picks up the finished boxes from manufacturing, stores them (in the Dispatch Warehouse) and hands them out to authorised collectors (when released by sales)
Sales	Sells both Target and Uncle Jim's brands Has a dedicated team and Account Manager for Dibbers Has a dedicated Business Unit for Web fulfilment
Web fulfilment	Handles sales orders and customer queries generated from the UncleJimsSweets.co.uk web site. This is a growing sales channel although not properly integrated into the rest of the company.
Marketing	Responsible for branding, package design, new products. Also handles PR
Compliance	Ensures that products meet International, National, and Corporate standards. Contains the Chemical Analysis team who check materials and products Performs internal Product Quality Assurance
R&D	Research and development acts as an "Internal Consultancy" to the rest of the company. This is a small department of scientists and their support team. R&D provide other departments with specialist advice as needed.
H&S	Health & Safety Ensures that employees work in a safe and healthy environment. Particularly interested in the ingredients used in production and the details of how such materials can be safely managed. Responsible for creating and maintaining OTGIs (On The Ground Instructions) given to staff concerning materials handling and storage
FM	Facilities Maintenance Cleaning and maintenance of infrastructure The FM Manager is also responsible for the maintenance of the manufacturing machinery (including fork lift trucks etc). This is provided by a mixture of dedicated CS&F employees and service contracts
IT	Handles IT systems, Help Desk, installation, upgrades and support. Controls ERP system and admin packages
Corporate Services	HR, Finance, Admin, Canteen Supervision (Canteen is self-financing but employs full time or part time staff

Assessment task 1: Elevator pitch (2 marks)

Imagine that you have been introduced to Alan Forden the MD (CEO) of CS&F. Alan is very keen to take up Lean 6 Sigma but is having difficulty explaining to his fellow Board members why it would be a good idea. Evidently, some members of the Board have expressed scepticism as they don't think that quality initiatives are capable of driving costs down in a significant way.

Alan has asked you to attend the next board meeting and provide a very short overview of what Lean 6 Sigma is and why CS&F should embrace Lean Thinking.

What you need to do:

Prepare a **short** (3 minutes maximum!) presentation explaining how embracing Lean 6 Sigma will be beneficial to CS&F and respond to the Board members concerns.

As an example of how simple and effective Lean Thinking can be, identify two simple tools which you could teach to CS&F employees that would help them develop a Lean viewpoint straight away (use tools discussed in the Yellow or Green Belt courses). Describe these in your presentation.

You will be given a chance to give this presentation during the first day of the course. Successful completion of this task will count towards your Black Belt assessment

Tip*: Keep this short and sweet – you only have 3 minutes to deliver your presentation. Black Belts need to be communicators and good communicators know their audience and adapt their presentations to fit. You audience is the Board of Directors of CS&F and the aim is to grab their attention and make them think that Lean thinking is a good philosophy to invest in.*

Assessment task 2: What's the question? (3 marks)

There is a concern that Production Line B is experiencing a lot of delays. As part of an initial investigation the production team where asked to log all breakdowns and delays with an estimation of how long the delay was. The log is reproduced on the next page

You have been asked to help prepare the initial Data Collection Plan

Using the Log on the next page see if you can provide answers to the following:

(1) What Research Questions should the team be asking?

(2) Can you spot any "Quick wins" or activities that the team could undertake while formal data collection is taking place?

Reason for stoppage	Delay time
Wrapper feed playing up	10 minutes
Wrapper mechanism jammed	25 minutes
Wrapper part replaced	13 minutes
Waiting for flavourings	14 minutes
Rap slit again line shut off	25 minutes
Auto wrapping problems	30 minutes
Dispenser funnel jammed	17minutes
Out of sugar	9 minutes
Plastic wrapping film split	21minutes
Sweets not cooled enough to wrap (switched off line, switch fan speed up)	4 minutes
Wrapper feed stuck	18 minutes
Sweet boiler unit cleaning	27 minutes 40 seconds
Product switch over (had to find cleaning tools)	3 hours
Wrapper unit belt fell off	6 minutes
Hopper overflowed	10 minutes
Wrapper belt snapped - replaced	14 minutes
Wrapper belt fell off - refit and adjust	8 minutes
All three heating units had to be cleaned as line had stopped B4	18 minutes
Sweet trolley nicked by thieving warehouse lot stopped us moving sweets	20 minutes
Wrong flavourings delivered	21 minutes
Motor problems	5 minutes
Wrapping cutter U/S - cleaned	17minutes
Out of bags	11 minutes
Wrapper guillotine stuck	16 minutes
Wrapper cutter adjusted	10 minutes
Lost OTGI (Recipe for mixture)	7 minutes
Sweets not wrapped right reset equipment	14 minutes
Essential maintenance to wrapper – electrician attends and replaces selector contact as burnt out and needed replacing quickly	31 minutes 20 seconds
Heater jammed needed TLC	30 minutes
Mike forgot to switch on mixer	5 minutes
Wrapping film problems	5 minutes 40 seconds
Fan overheating - replaced	49 minutes
Plastic wrapping film split	21 minutes
Plastic film spool feed clogged	27 minutes
Sugar container stuck so we could not use until we found a lever to get the top off	6 minutes

Assessment task 3: Training the Team (6 marks)

Black Belts perform training. In this assessment you will prepare and deliver a short training module on a basic Lean 6 Sigma topic or technique.

You will need to identify:

- Who the training is aimed at (this could be CS&F staff or personnel from your own organisation)
- What the topic is (agree this with your instructor)
- An exercise that the class could perform to reinforce your training (**Note:** all you need to do is describe the exercise)

Tips:

Your Instructor will set the duration but typically this is between 5-10 minutes so choose a simple topic and keep it brief.

Don't forget to introduce yourself and the topic

We are **not** looking for polished training material. Hand written notes, flipchart paper, basic slides are all acceptable

---

Sketch Your Training Plan here (intro, main body and exercise)

Assessment task 4: Measuring Capability (2 marks)

The Logistics division of CS&F is responsible for picking and delivering orders to their wholesale customers.

Delivery is scheduled for the “midday slot” the goal is to deliver at 12:00 (noon), however two hours on either side is considered acceptable.

In line with industry best practice CS&F look for a value of 1.33 for both Cp and Cpk as this is a good compromise between cost and performance.

The company have recorded process performance over a 5 day period in a Minitab Worksheet. You will use this data to determine how capable the process is.

Follow the steps through using Minitab then prepare a short description of the situation in terms that **Business would understand**. Perhaps you would wish to indicate if the process is satisfactory and if not how much trouble it is in.

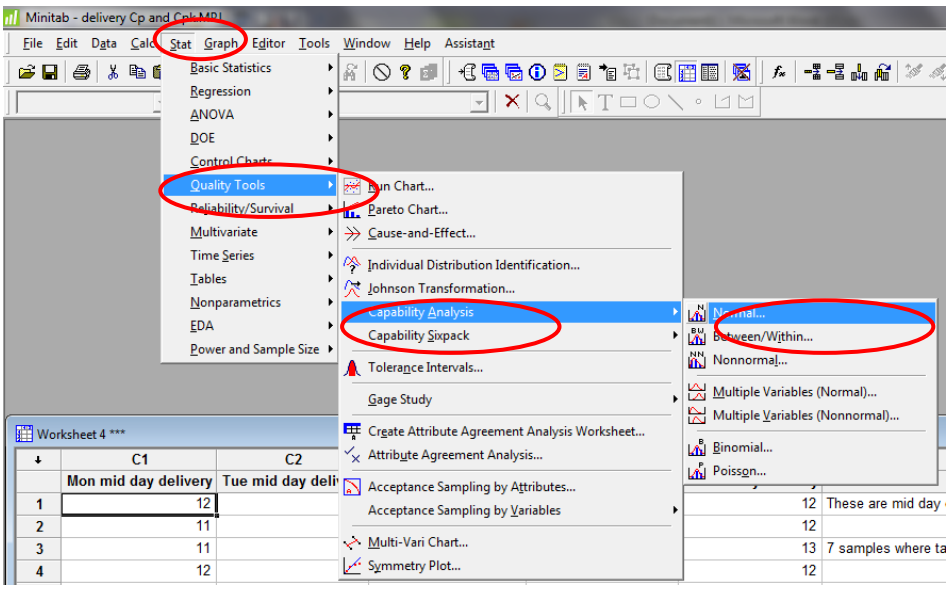
Step 1: Open file “delivery Cp and Cpk.MPJ” in Minitab

- Open a new project – (File ⇨ New ⇨ Project). If prompted: do NOT save your data
- Navigate to your course folder (File ⇨ Open ⇨ …)
- Open the file “Capability example”
- This pre-loads the worksheet for this exercise:
-

Capability example.mtw ***

↓	C1	C2	C3	C4	C5	C7-T
	Mon mid day delivery	Tue mid day delivery	Wed mid day delivery	Thu mid day delivery	Fri mid day delive	
1	12	10	13	13	12	These are mid day delivery times for our parcels
2	11	12	12	12	12	Missing
3	11	12	11	11	13	7 samples where taken each day - as shown
4	12	10	13	12	12	Missing
5	10	14	12	12	11	What are the cp and cpk values??
6	11	12	12	13	12	Missing
7	11	13	11	14	11	Is the process capable of delivering on time??

Step 2: Select the test you wish to run

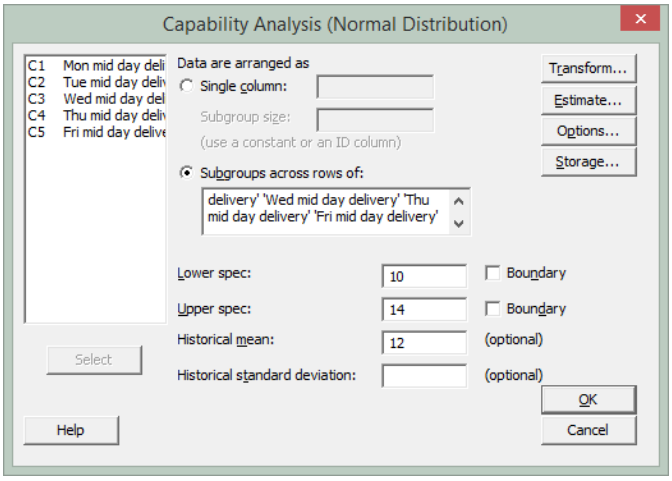


Step 3: Configure the test:

Enter all the columns (C1-C5) in the "Subgroups across rows of:" box.

To do this click the radio button by the heading then click on each column in the left hand box until it appears in the box under the heading.

Note: *your screen may look a little different from the screenshot as when you select a column it will move from the LH box to it's new home.*



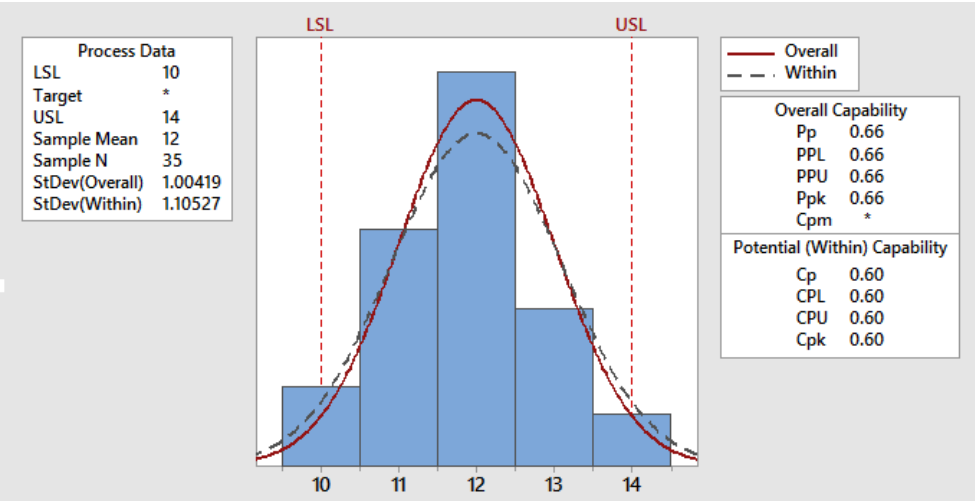
Set the Specification limits and mean as shown:

- Lower = 10 (or 10.00am in this case)
- Upper = 14 (or 2.00pm in this case)

The historical Mean is set to 12 (12.00 noon) which is the target time

When you are ready click on "OK" – Minitab will draw up a chart

Step 4: view the results and draw conclusions – is the process capable??



Like many stats programs, Minitab will provide lots more parameters than we need for this task. Concentrate on Cp and Cpk and create a brief report explaining in **business terms** what your findings are:

Assessment task 5: What next? (3 marks)

You have been approached by Libby Randall the Transport Manager for CS&F for some advice.

She has seen the evidence of the Paired t-test for consumption and does not know how to proceed.

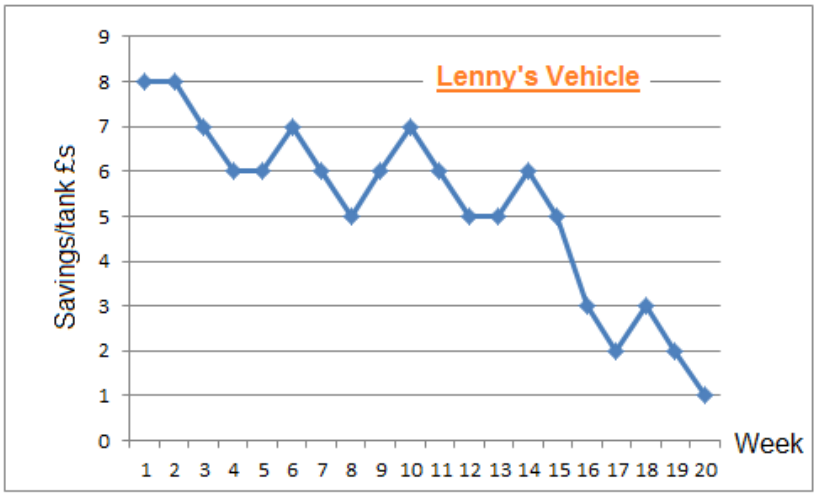
Currently each Lorry in the fleet is serviced once per your when it also has its yearly MOT (Road Safety Test) carried out. The Lorry manufacturer specifies two types of service a “Basic Service” and a “Full Service”. A Basic service consists of basics checks plus cleaning and replacement of the Air Filter. Every other year the Manufacturer specifies a “Full Service” which in addition to the Basic Service includes stripping down the engine, replacing the fuel injectors, and re-tuning. A full lorry service costs £400 plus parts (this is for 3 hours of a mechanics time @ £85/hour plus consumables). An MOT costs £120. A Basic Service costs £100 (an hour of a mechanics time @£85/hour plus consumables). New Lorries under 3 years old do not require the MOT and also any parts other than expendables are replaced free of charge under manufacturer’s warranty.

A cost for routinely fitted parts is shown below:

Part	**Costs**		**When fitted**
	Part	**Fitting**	
Front Brakes	£80/pair	£40	As needed
Rear Brakes	£100/pair	£80	As needed
Windscreen wash	£5	In Basic Cost	Topped up every service
Oil	Top Up £5 £95	In Basic Cost	Topped up every service Replaced on Full Service
Oil Filter	£35	£20	Changed on Full Service
Air filter	£3	In Basic Cost	Every Service
Tires checked and pumped up		In Basic Cost	Every service

Libby feels that after a service there is a drop in fuel consumption which saves between £5-£10 for each full tank of diesel, however the effect reduces sharply over time. The type of service (Full or Basic) does not seem to make a difference to the fuel savings. On average, each lorry uses about 8 tanks of fuel per week.

Libby has recorded how the extra savings per tank drop off calculating on a weekly basis over 20 weeks on a typical lorry:



Note: Vertical axis

This charts the savings per tank after Lenny had his truck serviced. So for the first two weeks after the service, each tank of fuel allowed the truck to travel a little further than before. To purchase fuel for the equivalent amount of miles would have cost the company £8.

Libby is keen to save money and help protect the environment but does not know what to do next? What would your advice be?

What you need to do:

Looking at the data that Libby has provided you with, and your own common sense, identify a cost effective action plan that could make a saving within the philosophy of Lean Thinking

Tips:

Think of "Quick wins" then areas that Libby might want to investigate and try and describe a long term vision of how Lean Thinking could be utilised to save fuel – remember Lean likes Low Tech simple cheap solutions (look to change the system not add more resources). Reflect on the Lean concept of "Empowerment"

Your Advice:

Assessment task 6: The right hypotheses tests (4 marks)

You have been asked to advise a Green Belt from the production team who is running a DMAIC project. The project is about to start the Measure phase and the Data Collection Plan is being developed.

The Green Belt wants your input so that the correct data is collected

What you need to do:

In this exercise you will focus on which Hypotheses tests could help answer the research questions of the team.

Read the description of the problem below, define (or refine) the Research Question(s) and then identify a suitable hypotheses test or tests which could help answer the questions.

Identify the type of data to collect and estimate the frequency and amount needed (eg "Continuous data, collect 4 sweets every two minutes for 30 days")

Wrapping Station

Manufacturing has problems with the wrapping machinery in both production lines. They are getting constant miss-wraps although there is a slight difference experienced when comparing the two lines with each other. The manufacturer of the wrapping station has sent over some information which allows the Green Belt to identify the average failure rate for a well maintained system. Manufacturing are keen to understand if there is a difference between the two lines and if they vary from the statistical norm. The GB has already checked that the data appears to be normally distributed.

Your Advice:

(1)What are the research questions to be answered?

(2) Hypotheses tests recommended

(3) Data to be collected

Tip:

Think about the type of data to be collected – remember the data type influences your choice of test

Assessment task 7: Control Charts and SPC (6 marks)

You have been asked to advise a Green Belt from the Web Fulfilment Team who is running a DMAIC project. The project is about to start the Measure phase and the Data Collection Plan is being developed.

The Green Belt wants your input so that the correct data is collected

What you need to do:

Read the description of the problem below, define (or refine) the Research Question(s) and then identify a selection of suitable SPC tools which could help answer the questions.

You must Include at least one Control Chart type or types and identify sample size, and probable number of samples plus appropriate measurement timescales. Also feel free to identify other SPC tools (such as Pie charts, Histograms/ Bar charts, Box and Whisker charts, Hypotheses tests) which might be useful.

You will need to identify at least two tools (Control Cart + one other) in this exercise

Box fill

The Web Fulfilment Team (WFT) has just introduced a Premium Package offering on the Uncle Jim's website. This consists of a gift wrapped box containing sweets, flowers and wine. The boxes need to be sturdy as well as carefully and neatly packed as the price to the customer is significant. As this is a new venture, the management team are keen to understand their performance both now and as time progresses.

From their general pool of packers they have selected a team of 4 experienced staff to handle these more complex orders. Each member has had special training. As a new venture, the team feel that the packing time is slower than it could be, however this is OK at the moment as the emphasis has to be on "getting it right" rather than speed. At the moment they are getting between 10-20 Premium orders each day they expect this to increase slowly to about 50. The packers have other duties but the premium packing task is their priority. At any time there could be between zero and the full team of 4 working on Premium orders.

The WFT Management team is keen to understand the time taken to pack each order, so that they can demonstrate improvement, share best practice and generally remove waste. They want to know how each individual packer performs and also how the group performs and they want to see the changes over time.

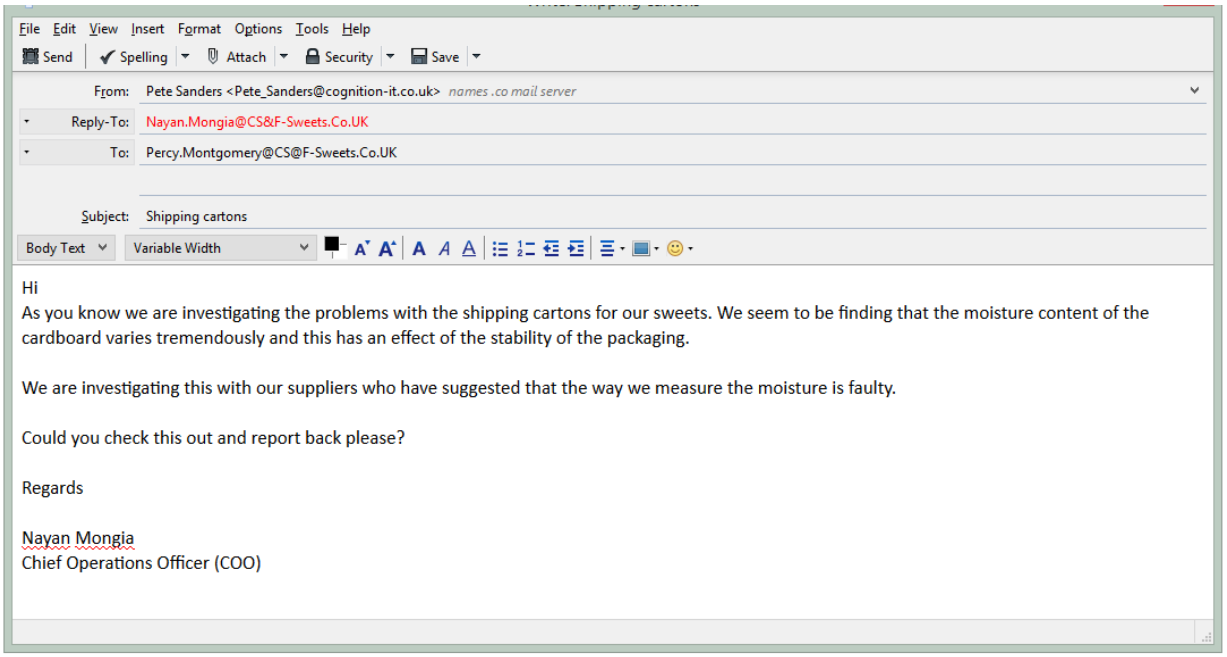
(Exercise continues over the page …)

Your Advice:

(1) What are the research questions to be answered?

(2) How would you model this process in a way to provide the understanding needed by the business? (at least 1 control chart plus parameters and 1 other SPC tool)

Assessment task 8: MSA (6 marks)



File Edit View Insert Format Options Tools Help

Send | Spelling | Attach | Security | Save

From: Pete Sanders <Pete_Sanders@cognition-it.co.uk> *names .co mail server*

Reply-To: Nayan.Mongia@CS&F-Sweets.Co.UK

To: Percy.Montgomery@CS@F-Sweets.Co.UK

Subject: Shipping cartons

Body Text | Variable Width

Hi

As you know we are investigating the problems with the shipping cartons for our sweets. We seem to be finding that the moisture content of the cardboard varies tremendously and this has an effect of the stability of the packaging.

We are investigating this with our suppliers who have suggested that the way we measure the moisture is faulty.

Could you check this out and report back please?

Regards

Nayan Mongia
Chief Operations Officer (COO)

Percy (Outbound Stores Manager) has asked your for help

Under your direction a representative sample consisting of 8 cartons was selected and two appraisers tested the moisture using a single moisture monitor. Each tester measured each carton three times.

The results are recorded in a Minitab worksheet – **Carton MSA data**

What you need to do:

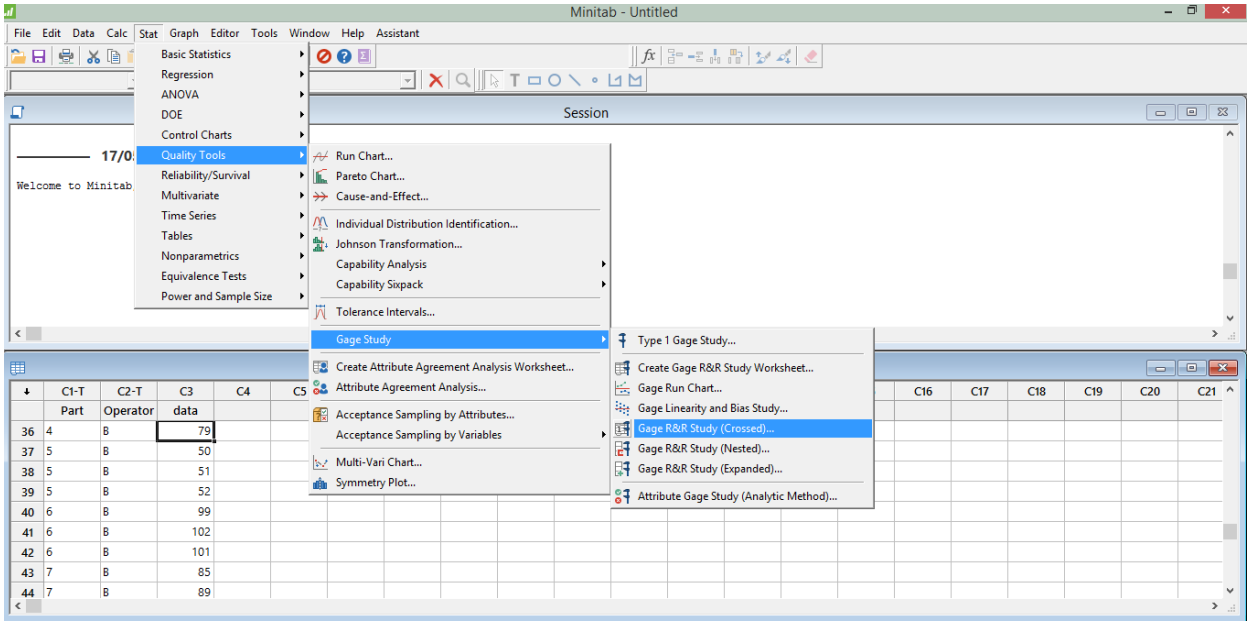
Use a Minitab's **Crossed Gauge Study** against the Carton MSA data, analyse the output then draft a report giving your opinion of whether the supplier is correct. Remember to justify your conclusions from the data.

NB: *The data is also available as an Excel spreadsheet under the same name and included as a print out over the next two pages*

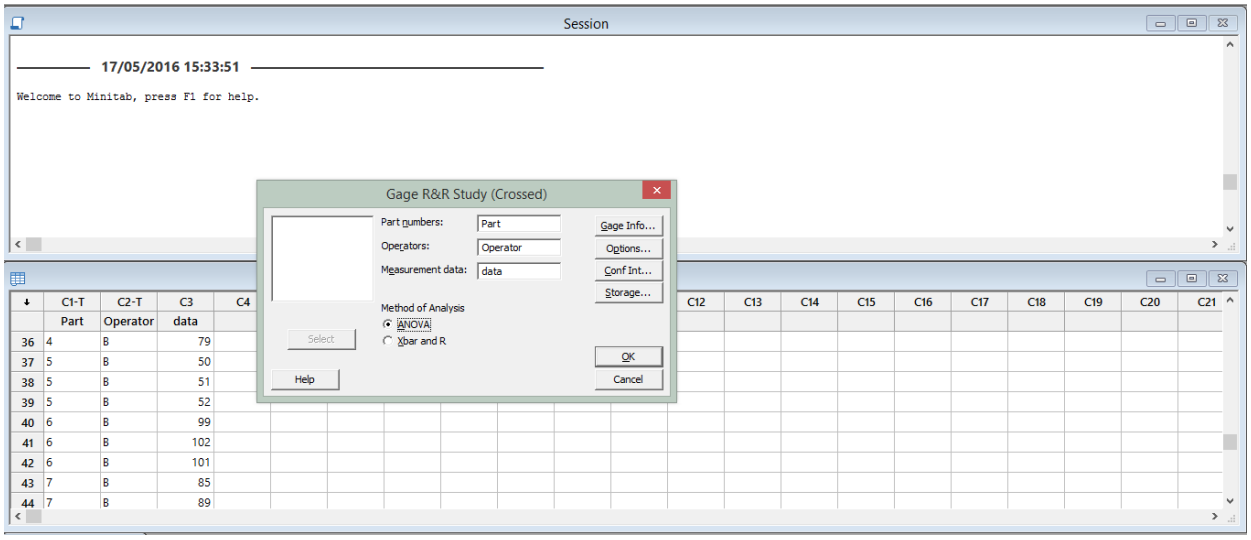
(Exercise continues over the page …)

Walk through:

(1) Find the test



(2) Configure the test



(3) Results

Results for: Carton MSA data.MTW

Gage R&R Study - ANOVA Method

Two-Way ANOVA Table With Interaction

Source	DF	SS	MS	F	P
Part	7	10352.7	1478.95	144.624	0.000
Operator	1	44.1	44.08	4.311	0.077
Part * Operator	7	71.6	10.23	2.134	0.068
Repeatability	32	153.3	4.79		
Total	47	10621.7			

α to remove interaction term = 0.05

Two-Way ANOVA Table Without Interaction

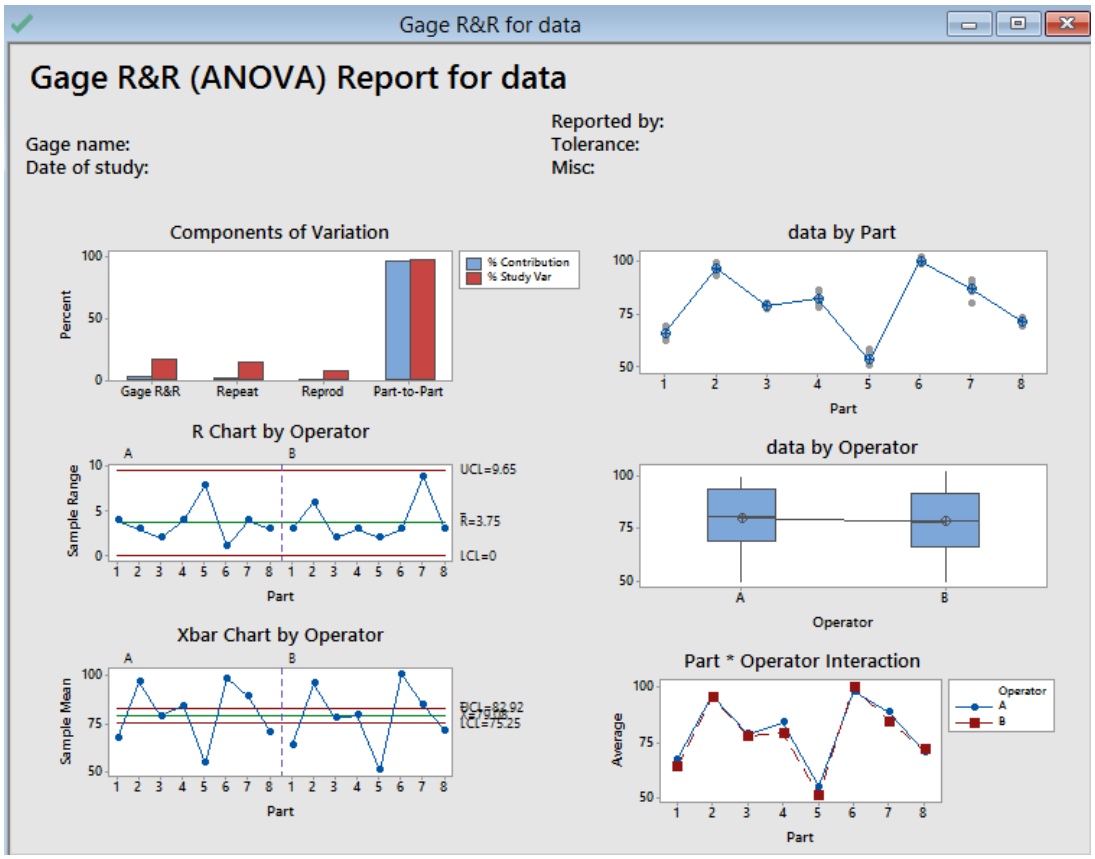
Source	DF	SS	MS	F	P
Part	7	10352.7	1478.95	256.447	0.000
Operator	1	44.1	44.08	7.644	0.009
Repeatability	39	224.9	5.77		
Total	47	10621.7			

Gage R&R

Source	VarComp	%Contribution (of VarComp)
Total Gage R&R	7.364	2.91
Repeatability	5.767	2.28
Reproducibility	1.597	0.63
Operator	1.597	0.63
Part-To-Part	245.531	97.09
Total Variation	252.894	100.00

Source	StdDev (SD)	Study Var (6 × SD)	%Study Var (%SV)
Total Gage R&R	2.7136	16.2816	17.06
Repeatability	2.4015	14.4089	15.10
Reproducibility	1.2635	7.5812	7.95
Operator	1.2635	7.5812	7.95
Part-To-Part	15.6694	94.0166	98.53
Total Variation	15.9027	95.4159	100.00

Number of Distinct Categories = 8



Report

Assessment Task 9: The right metrics (6 marks)

The Web Fulfilment Team (WFT) Green Belt who you assisted in Assessment Task 8 has returned to you asking for more advice, this time he needs help with choosing Lean Metrics. Here is the information again …

Box fill

The Web Fulfilment Team (WFT) has just introduced a Premium Package offering on the Uncle Jim's website. This consists of a gift wrapped box containing sweets, flowers and wine. The boxes need to be sturdy as well as carefully and neatly packed as the price to the customer is significant. As this is a new venture, the management team are keen to understand their performance both now and as time progresses.

From their general pool of packers they have selected a team of 4 experienced staff to handle these more complex orders. Each member has had special training. As a new venture, the team feel that the packing time is slower than it could be however this is OK at the moment as the emphasis has to be on "getting it right" rather than speed. At the moment they are getting between 10-20 Premium orders each day they expect this to increase slowly to about 50. The packers have other duties but the premium packing task is their priority. At any time there could be between zero and the full team of 4 working on Premium orders.

This is a High End product with a significant profit margin. Management consider that the VOC needs a "first class service" and that the **end-to-end service** must be perceived as "High Quality" otherwise customers will not re-order. The WFT Management team is keen to understand the time taken to pack each order, so that they can demonstrate improvement, share best practice and generally remove waste. You have already recommended a suite of SPC tools (Assessment Task 8), however the Management Team are now asking you to recommend some Metrics to help them further understand how the process works and how it meets the VOC. Customers can order from either the Website or by Phone. Delivery is by a specialist "next day delivery" service with the promise that all orders are received before 15:00hrs will be delivered the next working day. The flowers are provided by a local commercial nursery (Posh Plants) which specialises in high quality and exotic flowers. The flower part of the order is sent directly to Posh Plants if an order is placed on CS&F Website; if the order is received by phone, at present, the company uses a manual system were the operator who took the call sends an e-mail direct to Posh Plants as soon as the customer has hung up. Posh Plants deliver twice a day – once at 12:00hrs and again at 16:00hrs. The Delivery Service picks up at 14:00 and again at 18:00 if requested to (the last collection has to be specifically requested by phone or email). Orders picked up by 18:00 are guaranteed next working day delivery.

Tip: *Consider what the major concerns of the WFT Management might be. Consider where there is the possibility of error occurring in the process. Think about what makes a good metric, think of "Value For Money" and what message the metric will send. Consider negative impacts and using Secondary Metrics.*

End-to-End service *means the totality of the customer's experience – in this case it covers everything from the customer placing an order to the delivery of the package*

What you need to do:

Using your knowledge of Lean Metrics advise the GB on which would be most appropriate for the team to measure

Provide guidance on when, and how much, and what type of data needs to be collected

(Exercise continues over the page ….)

Your Advice:

What are the research questions to be answered?
Which metrics would help answer those questions?
How and in what way could the data be collected?

Assessment Task 10: Satisfactory Completion of Statistical Exercises (4 marks)

Identify four tests from group A and four tests from group B that you have satisfactorily performed during the week:

Group A

Anderson-Darling

I-MR

X BAR-R

U Chart

P Chart

Group B

1 sample t-test

2 sample t-test

Paired t-test

One-way ANOVA

Test for equal variance

2 proportion test

Chi-Square test

Assessment task 11: The Final Case Study (8 marks)

Senior Management is keen to evaluate the efficiency of the sweet production line. Previously the company has been to Gemba and investigated. What they found is recorded in the report following this text.

Alan wants you to focus on this as a priority. Management needs to understand the following:

- How much material is being lost (from supplier delivery to shipping)
- Where the bottle-necks are

Your task

You will be investigating Production Line A:

1. Draw up a Value Stream Map showing production from raw materials delivery to dispatch (6 blocks based on the section headings in the following description is sufficient).

   For each block show

   - First Time Yield

2. Calculate
   - OEE and TEPP – (When calculating availability use the production schedule provided in the text)
   - First Time Quality / Rolled Throughput Yield (RTY)
   - See if you can work out exactly how much of a tonne of sugar actually ends up as saleable sweets
   - Dock To Dock Time

3. Identify the production bottleneck

4. Identify any other metrics or Control Charts that would be useful in monitoring and understanding the process

5. Using the schedules provided calculate the **Build to Schedule** metric

6. Present your findings to the class

Report: The sweet production process @ 23rd April

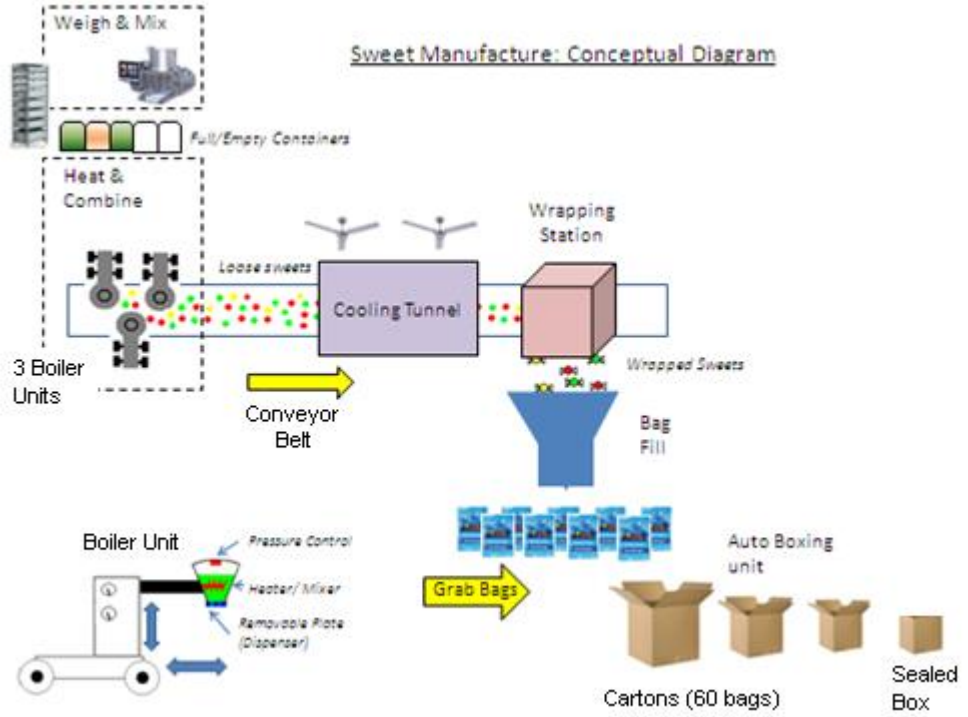
Pre-Manufacturing

Uncle Jim's produces and distributes their products from a central production facility. The raw materials for the sweets are obtained by the purchasing department and held in a central warehouse. The warehouse team keep a note of stock levels and when an item reaches a pre-determined figure they notify the purchasing department of the need to acquire another batch.

Manufacturing informs the warehouse every Friday of the ingredients needed for the following week. Over the weekend the requested quantities are assembled and shipped to the manufacturing holding bay early on Monday morning. The ingredients are tested (these tests only take about 30 minutes) and if satisfactory are used in manufacturing the sweets. The production lines consume about 10-12 tonnes of sugar each day (1 Metric Ton / tonne = 1,000,000 grams)

When the team visited the warehouse they found that the sugar was held in store for between 5-15 days (at this moment CS&F are not interested in the other ingredients). They also found that loss due to wastage was running at 20%.

The manufacturing process



The sweets are produced as continuous batches during the day. Shifts are 05:00 -13:00 and 12:00 - 20:00.

Production does not start until 06:00, the first shift actually arrives at 05:00 each morning which allows the operators to organise themselves, make sure the first ingredients are ready and start up the line. A mechanic is also in attendance so that if the operators find any problems beyond their capability a rapid fix can be assured. Production stops at 18:00, at this time the line is switched off and cleaned. Staff are paid for two hours but in practice the experienced teams usually only need one hour. Any serious faults are notified to maintenance who schedule repairs at the beginning of the morning shift (which sometimes starts late due to maintenance delays). During the day, as well as operations staff, there are two cleaners in attendance to keep the line clean. The shifts overlap in the middle of the day by 1 hour. This allows for an orderly handover and also for the early shift to tidy away and make good as necessary.

The company have been assured that the line should be capable of producing 80 cartons (60 packs of 100 sweets) each hour and are puzzled as to why this is not so.

Procedure	Time observed
Weigh and mix required ingredients and sugar	About 6 minutes per cycle
Heat and combine	20-30 minutes to get ready Then constant delivery
Cool and wrap	5 minutes
Bag and box	Complete box about 5 minutes

The capacity of a batch is constrained by the heating units which have limited capacity and take time to melt the sweet mixture. The heating units need to be fed constantly while in operation. There are three heating units which can be used each with an approximate throughput of 50 sweets/second. The maximum capacity of the wrapping unit is 250 sweets/second, the cooling 1,000 sweets/second and the bagging 800 sweets/second. The heating units need to be carefully managed as the sugar cannot be heated for too long without spoiling. Because of problems with the wrapping station causing the line to halt, heated mixtures often spoil resulting in further delay as the heating units are cleaned out and a new batch heated.

Weigh and Mix

Batches are created in 250ltr lots which is the optimum size needed to fill a sweet boiler in the Heat and Combine stage. The weigh and mix team use 20 containers which they refill as required once the contents have been consumed by the next stage (Heat and combine). The weigh and mix team have three mixer units, each unit can mix up to 75ltrs at a time. A cycle of weigh and mix takes roughly 6 minutes per machine although this varies considerably with the experience of the operators. Two operators are needed and depending on their experience can keep all three machines in continuous flow. They discovered that the operators working in Weigh and Mix were “busy” but able to satisfy the downstream demand with some time to spare. They discovered that sometimes the operators were distracted from weighing and mixing by the need to collect empty containers from Heat and Combine (The H&C operators are supposed to stack the used containers at a central point but often “forget”).

Production estimate that they waste about 5% of the ingredients at this stage over the course of a week (spillage occurs from time to time and mixed batches cannot be kept overnight so sometimes the last few containers of the day have to be thrown away). One litre of sweet mixture should make about 200 sweets before losses are taken into account. A sweet nominally weighs 5g, however there is considerable variation from operator to operator, according to the boiler operators this is inherent in their working practice and the crude machinery used (however, this variance adds to the “old fashioned charm” of the product in CS&Fs customers’ eyes). Staff in both the Weigh and Mix and the Heat and Combine cells use mechanical assists to move and position these heavy weights. The team found that occasionally there was a shortage of the mechanical assists as they were not returned by users.

Heat and Combine

Heat and combine is a continuous process as if a sugar boiler boils dry then the life of the heating coil is considerably shortened. Each production line has three boilers. At the beginning of the morning shift the boilers are filled and brought up to temperature. During the day they are topped up as required by their operators. At the end of the last shift they are left to cool down then cleaned thoroughly. The topping up process has been the focus of a previous kaizen which resulted in an OTGI defining a SOP. The “one best way” involves: releasing pressure on the boiler unit, adding in an additional 200lts of mixture, and then heating till the pressure builds up.

The investigating team endeavoured to identify the wastage from the boilers. The best they could come up with was that for every 200ltrs fed into a Boiling Unit (BU) about 180ltrs of sweet mixture was dropped on the conveyor belt - the rest of this sticky material adheres to the unit itself and needs to be cleaned off.

The three boilers are managed by a team of 2 operators. A good team can manage about 150 sweets/sec in total. The operators obtain the mixed material from the weighing station returning the empty cartons in the process. For a skilled team working with all three heating units; operating materials transfer takes about 15 minutes each hour, the SMED for charging a single boiler takes 4 minutes, cleaning and adjusting takes about 15 minutes. The team works for a full shift staggering breaks as they wish. They also keep their area clean and perform basic maintenance in line with Lean Cell philosophy.

Thanks to a quality improvement initiative, it only takes the company 30 minutes (of line downtime) to switch between brands, the boilers are the bottleneck here. At the end of the day the heating units need to be cleaned thoroughly, this takes 1 hour 30 minutes for each unit (1 hour to cool, 30 minutes to clean).

Cool and Wrap

The heated mixture drips onto the conveyor belt and flattens with impact. The combined output of all the heating units passes through a cooling tunnel for 4 minutes where filtered air is forced over them by fans. At the end of this process they pass into the wrapping station to be wrapped in plastic film.

The wrapping machinery is supposed to require 0.25 of a person to keep the plastic feed spools fed. Unfortunately the machinery is considered temperamental and constantly jams so it has been found necessary to have a full time team person there to manage it. The units are the oldest machinery on the line. The Senior Mechanic considers that the installation was incorrectly installed by the manufacturer, he feels that the work was incompetent and sloppy and that the machines have never lived up to the manufacturer's claims. He has access to the maintenance logs of other companies who use the same system and he says that this proves his point as they are much less troublesome. The wrapping machinery seems to have been gradually getting more and more temperamental and is now failing to correctly wrap about 15% of the sweets as well as jamming frequently. The Senior Mechanic has kept detailed records over the last few years which seems to prove the gradual degradation.

Sweets take about 4 minutes to cool. The cooled sweets are passed into the wrapping station where they are wrapped in plastic film. Hopefully a sweet is in and out of the wrapping station in about 15 seconds, unfortunately there seem to be problems with the machinery leading to unpredictable delays. On the day of the Gemba visit sweets could take anything from 15-45 seconds because of problems. The mechanics are at a loss to find a common cause for the problem; however one of the issues is that many sweets come out incorrectly wrapped. These sweets are thrown away. On the day measurements were taken approx. 15% of sweets were discarded in this way (the cooling tunnel is loss free). Management is concerned that the time taken to cool and wrap is slowly growing as the wrapping station becomes increasingly troublesome. This is having an effect on the productivity of the whole line. The Senior Mechanic says "that is what you get for buying rubbish in the first place".

The wrapping station specs state that each unit should be capable of wrapping 250 pieces per second, however during the Gemba visit it was noted that due to mechanical problems the wrapping station was off-line for about 20% of each hour!

Bag and Box

Sweets from the wrapping station are poured into the collection hoppers. This needs to be monitored by staff as sometimes the hoppers overflow. There are typically 5,000 sweets in the hoppers. The hoppers feed the feed mechanisms which drop batches of approx. 100 sweets into bags. The bags are then sealed and deposited into cartons. The equipment is the newest on the line and is fully automatic. The senior mechanic also has a low opinion of this machinery's manufacturer. A team member is needed about 0.25 of the time to make sure that there are enough cartons in the feed mechanism. Occasionally bags are not correctly sealed (about 1 in every 500).

The company have a dedicated staff member operating this unit, as well as managing the hoppers; the staff member identifies unsealed bags and removes them before they are boxed. Faulty bags are opened and the contents placed back in the hopper.

Filled cartons (each carton contains 60 bags of sweets) are removed from the line and placed on pallets waiting for removal to the warehouse. This occupies one team member 50% of the time (one person handles the bagging stations of both production lines). This part of the production line is very efficient and has virtually no loss (0.00001%).

Dispatch

The filled cartons are transferred to the dispatch warehouse and are delivered by a variety of methods. When measured it was found that filled cartons could stay at the end of the production line for up to 4 hours before being transferred to dispatch.

Typically cartons stay in the dispatch warehouse for up to 4 months before shipping.

There is some carton damage about 0.5% of cartons is found to be split and not fit to ship.

Production Line A Schedule (Planned): Monday 23rd April

Time	Activity	Output	Shift	Comments
05:00 – 06:00	Organisation and Preparation	nil	1	
06:00 – 12:00	Dibbers Mix recipe 1	360	1	
12-00 – 13:00	Switch over and handover	nil	1 & 2	Only 30 minutes down time expected Production stops 12:30 starts 13:00
13:00 – 16:00	Dibbers recipe 2	180	2	
16:00 – 16:30	Switch over	nil		Production ceases for 30 minutes
16:30 – 17:30	Uncle Jim's raspberry surprise	60	2	
17:30 – 18:00	Target raspberry shocks	30	2	No switchover as same sweet
18:00 – 20:00	Switch off and close down	nil	2	

Production Line A Schedule (Actual): Monday 23rd April

Time	Activity	Output	Shift	Comments
05:00 – 06:30	Organisation and Preparation	nil	1	Relay tripping, line started 30 minutes late
06:30 – 12:00	Dibbers Mix recipe 1	330	1	
12-00 – 13:00	Switch over and handover	nil	1 & 2	Only 30 minutes down time expected Production stops 12:30 starts 13:00
13:00 – 16:00	Dibbers recipe 2	180	2	
16:00 – 17:00	Switch over	nil		Production held up to service wrapping station
17:00 – 18:00	Uncle Jim's raspberry surprise	60	2	
18:00 – 20:00	Switch off amd close down	nil	2	

Note: *Output measured in cartons*

Your conclusions: